



Touchscreen phone and smartwatch with health sensor mobile app

all possible scenarios or operation modes in which there is a failure situation.

Pawar explains that, "As a medical device designer, one needs to be absolutely clear with two documents/standards, IEC 60601-1 and IEC 60601-1-2. If your device adheres to these documents, you would be what they usually call 601 compliant. The 601 standard is primarily for electrical safety and electromagnetic compatibility (EMC). If your medical device is using an external power supply, or even a step-down transformer, it needs to be 601-compliant as well. For EMC, testing needs to be rigorous to ensure that your device not only functions but also keeps functioning even with electromagnetic interference."

"There are two main safety approvals for power supplies. Information Technology Equipment (ITE) IEC 60950-1 and Medical Electrical Equipment (MEE) IEC 60601-1. There is also ISO 13485 certification, which states requirements for a comprehensive system of manufacturing of medical devices," says Chris

Jones, product marketing director of Artesyn Embedded Technologies. IEC 60601-1 is a safety standard for medical electrical equipment, whose third edition includes general requirements for safety as well as essential performance.

The software and firmware that goes in your medical device programming is also regulated. These are broadly classified by two testing approaches that are driven by simple code and complex code. For simple code, refer IEC 60601-1 Annex H, and for more complex code driven firmware, check IEC 62304 document.

"Making your medical device is a complicated and time-consuming process, and care should be taken to befriend these standards right from the prototype-design phase. Once the device is designed keeping standards in mind right from day one, the process becomes simpler, linear and cascading," adds Pawar.

Securing your body

Of course, with wireless access comes the possibility of hacking.

This could be especially painful now that the devices are on your body.

Barnaby Jack was someone who had already demonstrated how it was possible to hack a diabetic's insulin pump to deliver a fatal dose as well as how to hack a pacemaker from 15 metres away and get it to deliver an 830-volt jolt to the user. Unfortunately, he was found dead under suspicious circumstances a week before his scheduled demonstration at a major black hat convention with cyber security experts at Caesar's Palace. So, how can this problem be solved?

The knee-jerk response would be what former US vice president Dick Cheney's doctors did. They disabled his pacemaker's wireless connectivity to thwart possible assassination attempts. More thought-out approaches include mobile medical device benchmark initiative that was implemented by Centre for Internet Security, whose resulting benchmarks are recommended as guidance for device makers to harden a device's security.

Indeed, there are embedded cryptographic co-processors that come with processors like Cortex-M4 based STM32F479 to power applications requiring high security. A cryptographic co-processor is a hardware module specialised for encryption and related processing to prevent unauthorised retrieval of data. However, can chips like these solve the problem single-handed? Probably not. Jack had also added that many hospitals are using out-of-date software as they are afraid of running foul of regulations such as those formulated by the FDA. As a result, it is known that malware is rampant on hospital networks.

It looks like healthcare technology advances have gotten us to the stage where you do not need to worry that much about contracting a biological infection, but you do need to worry a lot about infecting yourself (and your devices) with malware. ●